MBO1 – MAXWELL BLOCK MINIMIZATION AND GAUSSIAN FLUCTUATIONS

Abstract. This focused working outline isolates the Abelian Maxwell block construction: a constrained minimizer, local fluctuation coordinates, exponential inverse-root locality and an exact Gaussian energy decomposition. It also records the photon domain estimates and exact cover identities needed later. Every overview box links to its statement and proof description.

Contents

1. Introduction and target statement 1
Proof overview 3
2. MB0. Cochains, averages and the two gauge choices 4
3. S03. The Landau minimizer and uniform differentiated locality 5
4. MBA. Axial coercivity and local fluctuation coordinates 6
5. M21. Exponential locality of a positive inverse square root 7
6. S04. The positive fluctuation covariance and exact Gaussian split 8
7. MBG. Curvature controls the Landau field in local real gauges 9
8. S05. Photon weakening, components and small-field domain margins 10
9. MBC. Consistency of the exact photon operators on odd covers 12

1. Introduction and target statement

MBO1 is the first multiscale proof project within the RGO1 stage of QEO1. The cochain averages and gauge choices are fixed explicitly. The matrix inverse-root lemma is stated independently of this lattice model.

Broader outlines. RGO1: renormalized construction and non-Gaussianity; QEO1: the full construction.

MBT. The Maxwell block theorem

Model-specific. Overview Target.

Setting. Fix an odd integer $L \geq 5$ and $0 < \alpha < 1$. At integer level $k \geq 0$ put $\eta = L^{-k}$ and use real periodic cochains on $\eta\mathbb{Z}^3/\ell\mathbb{Z}^3$, with pairing $\eta^3 \sum$ and mesh-normalized differences. The unit blocks are $\Delta_z = \{c : |c - z|_\infty < 1/2\}$, $z \in \mathbb{Z}^3/\ell\mathbb{Z}^3$. For $N = \eta^{-1}$ define the scalar and bond averages by

$$(\mathbf{Q}_k\lambda)(z) = \eta^3 \sum_{c\in\Delta_z} \lambda(c), \qquad (\mathbf{Q}_k\mathscr{A})_\mu(z) = \eta^3 \sum_{c\in\Delta_z} \sum_{j=0}^{N-1} \eta\mathscr{A}_\mu(c + j\eta e_\mu).$$

Write $\delta = d^T$, let $\mathcal{R}_k$ be the orthogonal projection onto $(\delta d)(\ker \mathbf{Q}_k)$ in scalar cochains, and define $H_k^{\mathrm{ph}}A$ by minimizing $\|d\mathscr{A}\|_2^2$ subject to $\mathbf{Q}_k\mathscr{A} = A$ and $\mathcal{R}_k\delta\mathscr{A} = 0$. Put $\Delta_k = (dH_k^{\mathrm{ph}})^T dH_k^{\mathrm{ph}}$ on unit bond fields.

For one block step, $B(y) = \{z : |z - y|_\infty < L/2\}$, $y \in L\mathbb{Z}^3/\ell\mathbb{Z}^3$, and

$$(\mathbf{Q}A)(y, y + Le_\mu) = L^{-4} \sum_{z\in B(y)} A(\Gamma_{z,z+Le_\mu}), \qquad (\tau A)(y, z) = \tfrac{1}{6} \sum_\pi A(\Gamma^\pi(y, z)).$$

The first paths are straight; the latter change coordinates in each of the six possible orders. All distances below are periodic sup distances of bond origins, or distances between the stated coordinate supports. A short Hölder difference compares parallel output bonds whose origins have distance in $(0,1]$.

Main statement. There are positive constants $C_P^{\mathrm{ph}}, \gamma_P$, depending only on L, α, with the following properties, uniformly in k and the admissible side. Here M is a power of L, $M \geq L^2$, and $\ell/(LM)$ is an odd integer at least 13.

The minimizer exists uniquely, is real linear and preserves constants. Its kernel, its first differences and its short Hölder first differences satisfy

$$|H_k^{\mathrm{ph}}(b,b')| + |\partial H_k^{\mathrm{ph}}(b,b')| + |\delta_\alpha \partial H_k^{\mathrm{ph}}((b,\widetilde{b});b')| \leq 3C_P^{\mathrm{ph}} e^{-\gamma_P d(b,b')}, \qquad |\Delta_k(b,b')| \leq C_P^{\mathrm{ph}} e^{-\gamma_P d(b,b')}.$$

There are local real coordinates V_k for $\{Z : \mathbf{Q}Z = 0, \tau Z = 0\}$, with a bijection C and $\dim V_k = (2 - 2L^{-3})\ell^3$, such that

$$\widetilde{\Delta}_k = C^T \Delta_k C > 0, \qquad |\widetilde{\Delta}_k^{-1/2}(\Upsilon, \Upsilon')| \leq C_P^{\mathrm{ph}} e^{-\gamma_P d(\Upsilon,\Upsilon')}.$$

For a real coarse field A' on side ℓ/L, let $A'_L(b) = L^{-1/2} A'(b/L)$ and let $A_k^{\min}(A')$ minimize $\langle A, \Delta_k A\rangle$ on $\{A : \mathbf{Q}A = A'_L, \tau A = 0\}$. With $\mathrm{Sq}_k = C\widetilde{\Delta}_k^{-1/2}$, the exact identity is

$$\langle A_k^{\min}(A') + \mathrm{Sq}_k\zeta, \Delta_k(A_k^{\min}(A') + \mathrm{Sq}_k\zeta)\rangle = \|dH_{k+1}^{\mathrm{ph}} A'\|_2^2 + \|\zeta\|_2^2.$$

The harmonic part of A' is arbitrary. The fluctuation law in these coordinates is the standard product Gaussian. Restricting it to $|\zeta(\Upsilon)| \leq p_0$, $p_0 \geq 1$, has mass $\exp(-\ell^3 \delta\epsilon_k^0)$ with $0 \leq \delta\epsilon_k^0 \leq 12e^{-p_0^2/2}$.

For each partition M-cube $\square$, let $\square^\dagger$ have the same centre and side $(2L+1)M$, as in MBG. There are also local real gauges with $\|H_k^{\mathrm{ph}} A - d\nu\|_{i,\square^\dagger} \leq C_B \|dA\|_\infty$, $i = 1,2,3$, where the norms are sup, first difference and short Hölder first difference and $C_B = C_B(L,M,\alpha)$. For $0 < \alpha_0 \leq 1/4$ and $804\alpha_0 \log M \leq \gamma_P M$, the explicit polynomial weakening of S05 preserves exponential locality on $|s_a| \leq M^{\alpha_0}$, with rate $\gamma_P/2$ and constant $116 C_P^{\mathrm{ph}} M^{402\alpha_0}$. Exact operators and exact block minimizers commute with the odd periodic pullbacks of MBC.

Proof work. The three substantial mechanisms are the alias calculation with its zero-alias cancellation, axial coercivity in local coordinates, and the weighted resolvent argument for the positive inverse square root. Path weakening and local gauges then give the photon inputs required by the nonlinear integration. The separate small-charge domain result is stated with its full numerical hypotheses in S05.

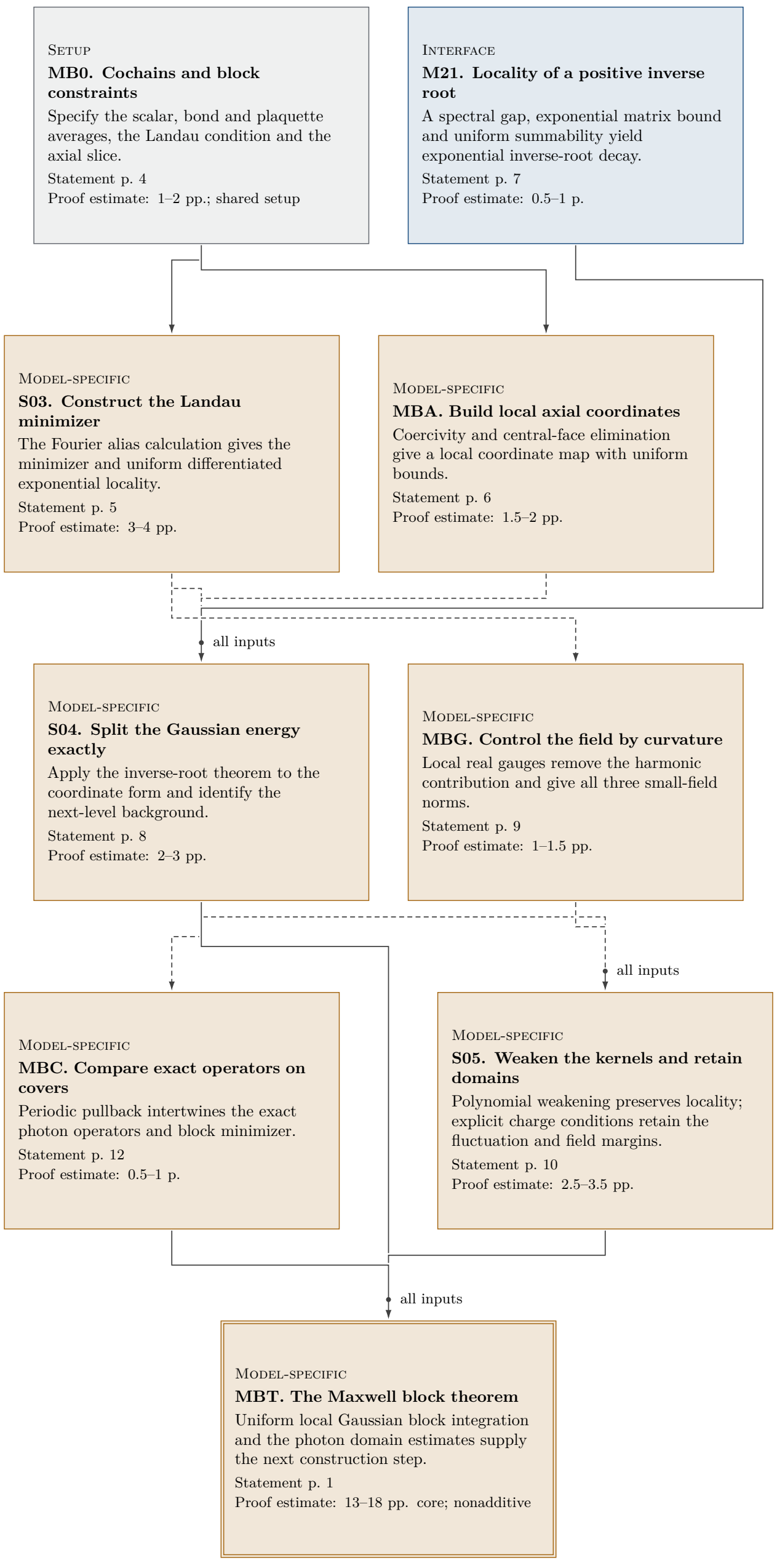
Inputs. Only the cochain scheme above and finite-dimensional Fourier, minimization and Gaussian algebra. No chamber fermion operator, mass tuning or interacting estimate is assumed.

Source. QEM1 §10, lines 7809–9071, pp. 89–102. Full write-up estimate: 13–18 pages, including shared definitions; supporting estimates below must not be added again to this total.

MBO1 – Maxwell block minimization and Gaussian fluctuations

Definitions/setup | Model-specific | Interface (general)

Focused outline under RGO1, within QEO1. The shared cochain scheme is MB0. Blue denotes the general inverse-root theorem; ochre denotes statements for this particular Maxwell blocking scheme.



Solid arrows supply inputs to an application; dashed arrows identify use inside a proof. Every incoming branch at a junction is required together with the standing hypotheses on the statement page.
Click a box for its statement. Return links are navigation, not proof dependencies.

All inputs at a junction are required. The domain conclusion S05 includes its own explicit charge inequalities. Exact cover consistency concerns unweakened operators. This is a one-step construction; the later iterated terminal-measure argument is a separate project. Proof estimates share setup and are not additive.

2. MB0. Cochains, averages and the two gauge choices

Setup. Overview Target.

Objects and hypotheses. Use the periodic three-dimensional cochains and pairings stated in MBT. The exterior difference is $(d\mathscr{A})_{\mu\nu} = \partial_\mu \mathscr{A}_\nu - \partial_\nu \mathscr{A}_\mu$, with $\partial_\mu = \eta^{-1}(\text{forward shift} - 1)$. On plaquettes the block average is

$$(\mathbf{Q}_k^{(2)} F)_{\mu\nu}(z) = \eta^3 \sum_{c \in \Delta_z} \sum_{i,j=0}^{N-1} \eta^2 F_{\mu\nu}(c + i\eta e_\mu + j\eta e_\nu).$$

The one-step unit-lattice averages and axial path functional are those displayed in MBT. Enlarged M-cubes have side $(2L+1)M$.

Definition and elementary identities. The Landau choice is the constrained curvature minimizer using $\mathcal{R}_k$; the axial choice fixes $\tau A = 0$ at one block step. The averages preserve constants and bond means, are onto on bonds, and satisfy

$$\mathbf{Q}_k d = d\mathbf{Q}_k, \quad d\mathbf{Q}_k = \mathbf{Q}_k^{(2)} d, \quad \|\mathbf{Q}_k^{(2)} F\|_2 \le \|F\|_2, \quad \mathbf{Q}_k \mathbf{Q}_k^T = 1 \quad \text{on scalars.}$$

The two kinds of averages live on different mesh levels and must not be identified by notation alone.

Proof work. Telescoping proves the differential identities. Jensen and counting give contraction. Prescribe boundary fine bonds to prove onto-ness. These checks remain within the setup argument rather than separate theorem boxes.

Source. QEM1:7822–7877,7949–7958. Setup budget: 1–2 pages, included in the group estimate.

3. S03. The Landau minimizer and uniform differentiated locality

Model-specific. Overview Target.

Objects and hypotheses. For odd $L \geq 5$, integer $k \geq 0$, $\eta = L^{-k}$ and integer side ℓ divisible by L, use exactly the averages and Landau constraint of MB0. Fix $0 < \alpha < 1$.

Claim. H_k^{ph} exists uniquely, is real linear, and $\mathbf{Q}_k H_k^{\mathrm{ph}} = 1$. It preserves constants and closed fields and is the identity at $k = 0$. Its Euler characterization is

$$\mathbf{Q}_k \mathscr{A} = A, \quad \delta d \mathscr{A} \in \operatorname{ran} \mathbf{Q}_k^T \text{ on bonds}, \quad (\delta d)\delta \mathscr{A} \in \operatorname{ran} \mathbf{Q}_k^T \text{ on scalars}.$$

There are $C_H(\alpha), C_\Delta, C_\Delta^{(2)}, \gamma_H > 0$, independent of k, ℓ, such that each of H_k^{ph}, $\partial H_k^{\mathrm{ph}}$ and $\delta_\alpha \partial H_k^{\mathrm{ph}}$ is bounded in absolute value by $C_H(\alpha) e^{-\gamma_H d(b,b')}$, and

$$|\Delta_k(b, b')| \leq C_\Delta e^{-\gamma_H d(b,b')}, \qquad \|\Delta_k\|_{2\to 2} \leq C_\Delta^{(2)}.$$

Here the Hölder output bonds are parallel, at distance in $(0, 1]$; the column and derivative direction are held fixed.

Proof work. First identify all unconstrained minimizers as one field plus $d(\ker \mathbf{Q}_k)$; scalar Laplacian injectivity fixes the Landau representative. For locality, separate the zero alias in the explicit Fourier system. The apparent pole cancels before taking $k \to \infty$. Bound the remaining alias sums in a fixed complex strip, keep the denominators uniformly away from zero, and shift the periodic Fourier contour. The summable differentiated alias bound uses $\alpha < 1$.

Inputs. The explicitly stated averaging scheme, periodic Hodge/Fourier algebra and finite-dimensional minimization. No inverse-root lemma is needed for this minimizer theorem.

Source. QEM1 `prop:photon-Landau-minimizer`,7884–7945; symbol/regularity proof 8002–8218, especially `eq:photon-zero-alias`,8062–8066, and `eq:photon-alias-Holder-sum`,8155–8159. Proof estimate: 3–4 pages beyond setup.

4. MBA. Axial coercivity and local fluctuation coordinates

Model-specific. Overview Target.

Objects and hypotheses. On the unit torus, with L-blocks and $\mathbf{Q}, \tau$ from MB0, let $\mathcal{Z} = \{Z : \mathbf{Q}Z = 0, \tau Z = 0\}$. In each block choose a translated orthonormal basis of the interior bond space $\ker \tau_y$, simultaneously diagonalizing its three coordinate reflections. Add the point masses on every noncentral face bond. Their orthogonal direct sum is V_k; its basis labels are Υ. A central face bond is the one on the line joining the two block centers.

Claim. For axial fields,

$$\|Z\|_\infty \le \tfrac{3}{2} L^2 \|dZ\|_\infty, \qquad \|Z\|_2^2 \le 9L^7 \|dZ\|_2^2 \quad (Z \in \mathcal{Z}).$$

Let $Cz = z$ on the retained bonds. On the central bond b_c of a face c, set

$$(Cz)(b_c) = -\sum_{b \in c,\, b \neq b_c} z(b) - L^{-1} \sum_{b \text{ interior}} n_c(b) z(b),$$

where $n_c(b)$ counts the straight averaging paths crossing that bond. This is a real bijection $C : V_k \to \mathcal{Z}$, and

$$\dim V_k = (2 - 2L^{-3})\ell^3, \quad \|z\|_2 \le \|Cz\|_2 \le \sqrt{5L^3}\|z\|_2, \quad \|Cz\|_\infty \le 3L^{9/2} \sup_\Upsilon |z(\Upsilon)|.$$

Each column satisfies $\sum_b |(C\Upsilon)(b)| \le 14L^{3/2}$, and lies within distance L of $\operatorname{supp} \Upsilon$. Coordinate sign changes and the relevant block translations permute the signed basis. All these constants depend only on L.

Proof work. For interior bonds, compare the six axial paths and use their strip curvatures. Use the coarse average to bound face means and transverse face differences. Count these local inequalities in L^2. The central-bond equations are independent and eliminate exactly the remaining block-average constraints; retain their coordinate counts and column supports for the covariance theorem.

Source. QEM1:7947–8000,8228–8321; in particular `eq:axial-fluctuation-coercivity` and `eq:photon-coordinate-bijection`. Proof estimate: 1.5–2 pages.

5. M21. Exponential locality of a positive inverse square root

Interface. Overview Target.

Objects and hypotheses. Let I be finite, let d be symmetric and nonnegative with $d(a,a)=0$, and assume $d(a,c)\le d(a,b)+d(b,c)+\phi$, with $\phi\ge 0$. Let T be real symmetric, $T\ge t_0 I>0$, and $|T(a,b)|\le C_T e^{-g d(a,b)}$ with $g>0$. Choose a positive number

$$S\ge e^{g\phi/2}\sup_a\sum_b (d(a,b)+\phi)e^{-g d(a,b)/2},\qquad \theta=\min\{g/2,\,t_0/(2C_T S)\}.$$

Claim. For $x\ge 0$,

$$|(T+xI)^{-1}(a,c)|\le (t_0/2+x)^{-1}e^{-\theta d(a,c)},\qquad |T^{-1/2}(a,c)|\le \sqrt{2/t_0}\,e^{-\theta d(a,c)}.$$

Uniformity over a family requires common t_0, C_T, g, ϕ, S; finiteness of each matrix alone is not a uniform summability bound.

Proof work. Conjugate by $e^{\theta d(\cdot,c)}$. Bound its perturbation of T by $t_0/2$ using the Schur test, then integrate the resolvent:

$$T^{-1/2}=\pi^{-1}\int_0^\infty x^{-1/2}(T+xI)^{-1}\,dx.$$

No photon or gauge hypothesis enters this finite-matrix lemma.

Source. QEM1:8374–8410, `eq:photon-root-integral`. Proof estimate: 0.5–1 page, shared with its application rather than a separate manuscript project.

6. S04. The positive fluctuation covariance and exact Gaussian split

Model-specific. Overview Target.

Objects and hypotheses. Use the concrete minimizer/energy form of S03 and the coordinates $C : V_k \to \mathcal{Z}$ of MBA. Put $\widetilde{\Delta}_k = C^T \Delta_k C$ and $C_k = \widetilde{\Delta}_k^{-1}$.

Claim. With $c_1 = 14L^{3/2}$ and

$$c_P = \max\{9L^7, 5L^3 C_\Delta^{(2)}\}, \qquad C_{\mathrm{Dt}} = c_1^2 C_\Delta e^{2\gamma_H L},$$

one has

$$c_P^{-1} I \le \widetilde{\Delta}_k \le c_P I, \qquad |\widetilde{\Delta}_k(\Upsilon, \Upsilon')| \le C_{\mathrm{Dt}} e^{-\gamma_H d(\Upsilon,\Upsilon')}.$$

Support distance has triangle defect at most L. Its uniform basis sum obeys

$$\sup_{\Upsilon} \sum_{\Upsilon'} e^{-g d(\Upsilon,\Upsilon')} \le c_{\mathrm{bas}}(g) := 2L^3 e^{2gL}(1 + 6/g)^3.$$

Set

$$S_P = e^{\gamma_H L/2}(L + 2/\gamma_H) c_{\mathrm{bas}}(\gamma_H/4), \quad \gamma_P = \min\{\gamma_H/2, (2c_P C_{\mathrm{Dt}} S_P)^{-1}\},$$

and $C_P^{\mathrm{ph}} = \max\{C_H(\alpha), C_{\mathrm{Dt}}, \sqrt{2c_P}, 4e^2\}$. Then M21 gives $|C_k^{1/2}(\Upsilon, \Upsilon')| \le C_P^{\mathrm{ph}} e^{-\gamma_P d(\Upsilon,\Upsilon')}$, uniformly in mesh and volume.

The constrained minimum $A_k^{\min}(A')$ exists uniquely. For $\mathrm{Sq}_k = C C_k^{1/2}$,

$$\langle A_k^{\min} + \mathrm{Sq}_k \zeta, \Delta_k(A_k^{\min} + \mathrm{Sq}_k \zeta)\rangle = \langle A_k^{\min}, \Delta_k A_k^{\min}\rangle + \|\zeta\|_2^2.$$

Moreover, for a unique real scalar ω_k with zero $(k+1)$-step average,

$$H_k^{\mathrm{ph}} A_k^{\min}(A') = (H_{k+1}^{\mathrm{ph}} A')_L - d\omega_k(A'), \qquad \|dH_k^{\mathrm{ph}} A_k^{\min}(A')\|_2 = \|dH_{k+1}^{\mathrm{ph}} A'\|_2.$$

For the standard Gaussian μ_I on V_k and $p_0 \ge 1$,

$$\mu_I\{\sup_{\Upsilon} |\zeta(\Upsilon)| \le p_0\} = e^{-\ell^3 \delta\epsilon_k^0}, \quad \delta\epsilon_k^0 = -(2 - 2L^{-3}) \log \Pr(|g| \le p_0) \le 12 e^{-p_0^2/2}.$$

Here g is a standard real normal variable. This is a one-step identity. An iterated photon tower and its global Jacobian are not conclusions of this card.

Proof work. Transfer axial coercivity through $\mathbf{Q}_k^{(2)} dH_k^{\mathrm{ph}} = d$ to bound the coordinate form below; use the local columns to bound it above and off diagonal. Verify every hypothesis of M21. Orthogonality to the fluctuation slice proves the energy split. Compare the two constrained minimizing problems to identify the next-level background up to its real zero-average gauge.

Source. QEM1, lines 8312–8350 and 8352–8535; exact split `eq:photon-Gaussian-block-decomposition`, lines 8478–8486; background identity 8492–8535. Proof estimate: 2–3 pages beyond the coordinate and matrix-lemma inputs.

7. MBG. Curvature controls the Landau field in local real gauges

Model-specific. Overview Target.

Objects and hypotheses. Let $M = L^m \geq L^2$, $\ell/(LM)$ odd and at least 13. Enlarge each partition M-cube $\square$ by LM in every direction to $\square^\dagger$, of side $s_d = (2L+1)M \leq 3\ell/13$. For fine photon coefficients let $\|\cdot\|_1, \|\cdot\|_2, \|\cdot\|_3$ mean sup, first difference and short Hölder first difference; local norms retain only contained bonds and difference cells.

Claim. For every real unit bond field A and every $\square$ there is a real scalar ν on its nonwrapping enlargement such that

$$\|H_k^{\mathrm{ph}} A - d\nu\|_{i,\square^\dagger} \leq C_B \|dA\|_\infty, \qquad i = 1, 2, 3.$$

The constant depends on L, M, α, not on k, ℓ or the harmonic part of A.

Proof work. Subtract the mean and gauge by ordered path sums in a lifted period cell. Inner bonds cost their distance times $\|dA\|_\infty$; the seam costs at most $3\ell^2\|dA\|_\infty/2$. Convolving with the differentiated exponentially decaying kernel absorbs that seam cost because the enlarged cube stays away from it. Remove the remaining constant field by a real local gauge.

Inputs. Only the concrete minimizer and its three uniform kernel estimates from S03, with the stated nonwrapping geometry.

Source. QEM1 `prop:photon-curvature-local-regularity`,8553–8645. Proof estimate: 1–1.5 pages.

8. S05. Photon weakening, components and small-field domain margins

Model-specific. Overview Target.

Objects and hypotheses. Use the photon kernels of S03 and S04, with common bound $C_P^{\mathrm{ph}} e^{-\gamma_P d}$. Parameter cubes have half-side $a = M$ or LM, centered at $a\mathbb{Z}^3/\ell\mathbb{Z}^3$; unlike polymer cubes they overlap. Cubes are linked when they share a lattice site. Let $\mathrm{Cub}_a(S)$ be all cubes meeting S and choose shortest rectilinear paths in coordinate order 1,2,3. The odd side fixes signed offsets uniquely. Smooth only the reconstruction row with the source partition χ_c, where

$$\vartheta(r) = 0 \ (r \le 0), \quad 3r^2 - 2r^3 \ (0 \le r \le 1), \quad 1 \ (r \ge 1), \qquad \chi_c(x) = \prod_{\nu=1}^{3} \vartheta(2(3/4 - |x_\nu - c_\nu|/M)).$$

For a bond b, set $\chi_c(b) = \chi_c(x_b)$, where x_b is its midpoint. Its sum is one and its first differences and their increments have the bounds $3/M$, $42|r|/M^2$ used in the source product rule. For a unit column bond b' with ends p_1', p_2', put

$$P'(c,b') = \mathrm{Cub}_M(\mathrm{Pth}(c,p_1') \cup \mathrm{Pth}(c,p_2')), \quad w(b,b';s) = \sum_c \chi_c(b) \prod_{a \in P'(c,b')} s_a.$$

For an interior coordinate let $\mathrm{St}(\Upsilon) = B(y)$, $\mathrm{Pt}(\Upsilon) = \{y\}$; for a face coordinate both are the two bond ends. Let $P''(\Upsilon,\Upsilon')$ contain every LM parameter cube meeting these two sets St or any ordered path between their points Pt. Define

$$H_k^{\mathrm{ph}}(s)(b,b') = H_k^{\mathrm{ph}}(b,b') w(b,b';s), \quad C_k^{1/2}(s)(\Upsilon,\Upsilon') = C_k^{1/2}(\Upsilon,\Upsilon') \prod_{a \in P''} s_a.$$

At $k = 0$ replace the first expression by $\delta_{bb'} \prod_{a \in \pi_M : b \subset a} s_a$.

Locality claim. If $0 < \alpha_0 \le 1/4$ and $804\alpha_0 \log M \le \gamma_P M$, then on the closed polydiscs $|s_a| \le M^{\alpha_0}$ the minimizer's three kernels and the root kernel have bound

$$116 C_P^{\mathrm{ph}} M^{402\alpha_0} e^{-(\gamma_P/2)d}.$$

Entries are polynomials, real for real parameters, and recover the exact operators at $s = 1$. When parameters vanish outside an active union, nonzero rows and columns stay in one linked component and read only that component's parameters. The exact symmetries are coordinate sign changes and the appropriate M or LM translations. The weakened root is not asserted positive or a square root away from $s = 1$.

Domain claim and its additional hypotheses. Fix the logarithmic cutoff exponents $p, p_0 > 0$. Let $0 < \varepsilon \le 1/200$, $e_k \in (0,1)$, and $t_{i,k} = e_k^{-3/4+i\varepsilon}$, $i = 1,2,3$. R_k is the complex field with all three norms below $t_{i,k}$. A field lies in $c\widetilde{R}_k$ if it is a global summand in cR_k plus a real field admitting independent real gauges on each $\square^\dagger$ with all three norms below $ct_{i,k}$. For $\widetilde{R}_k^-$ use local thresholds $t_{i,k}$ and global thresholds $(1 - e_k^\varepsilon)t_{i,k}$. Define

$$\mathrm{Dom}_k = \{A \text{ real unit field} : H_k^{\mathrm{ph}} A \in \tfrac{1}{4}\widetilde{R}_k\}, \quad p_{0,k} = (-\log e_k)^{p_0}, \quad p_k = (-\log e_k)^p.$$

Take $C_w \ge 116 C_P^{\mathrm{ph}} M^{402\alpha_0}$ and $0 < \gamma_w \le \gamma_P/2$. Let c_γ bound the two sums $\sup_\Upsilon \sum_{\Upsilon'} e^{-\gamma_w d}$ and $\sup_b \sum_{b'} e^{-\gamma_w d}$, uniformly in mesh/side. With $c_C = 3L^{9/2}$ require

$$p_{0,k} \le e_k^{-\varepsilon/4}, \quad 2C_w^2 c_\gamma^2 c_C e_k^{\varepsilon/4} \le 1, \quad 8C_B p_k \le t_{3,k}, \quad 8e_k^{3/4-7\varepsilon/2} \le 1, \quad L^{3/4} \ge 8/3.$$

Then $\|dA\|_\infty \le p_k$ implies $A \in \mathrm{Dom}_k$. Every ζ with $|\zeta(\Upsilon)| \le p_{0,k}$ gives

$$\mathscr{Z}_k(s,\zeta) = H_k^{\mathrm{ph}}(s) C C_k^{1/2}(s)\zeta, \qquad \max_i \|\mathscr{Z}_k(s,\zeta)\|_i \le \tfrac{1}{2} e_k^{-\varepsilon/2}.$$

For $A \in \widetilde{R}_k^-$, $A + t\mathscr{Z}_k(s,\zeta) \in \widetilde{R}_k$ on $|t| \le \frac{1}{2} e_k^{-3/4+9\varepsilon/2}$. If ℓ/L is also admissible and $e_{k+1} = \sqrt{L} e_k$, then $A_k^{\min}(A') + \mathrm{Sq}_k\zeta \in \mathrm{Dom}_k$ for $A' \in \mathrm{Dom}_{k+1}$.

Proof work. Bound each parameter path by $402(1 + d/M)$ factors and absorb their growth using half the exact exponential rate. Differentiate the row partition explicitly. For domains, sum the two photon kernels through the local coordinate map, then retain the source quarter-domain margin under the exact background identity and scaling.
Inputs and boundary. Only photon bounds and MBG are required here. A photon majorant can be fixed independently; choosing the same M, rates and charge ceiling for fermions later is an additional common-parameter task.
Source. QEM1:8647–9009; `prop:photon-weakened-bounds`,8773–8834, and `prop:photon-domain-preservation`,8924–8995. Proof estimate: 2.5–3.5 pages beyond its explicit kernel inputs.

9. MBC. Consistency of the exact photon operators on odd covers

Model-specific. Overview Target.

Objects and hypotheses. Let $\ell_2 = J\ell_1$, with J odd and both sides admissible, and use the same translated reflection eigenbasis in each block. pj^* is periodic pullback, including on the coarse lattices and coordinate spaces. Its inner products scale by J^3.

Claim.

$$O^{(2)}\mathrm{pj}^* = \mathrm{pj}^* O^{(1)}, \quad O = H_k^{\mathrm{ph}}, \Delta_k, C, \widetilde{\Delta}_k, (\widetilde{\Delta}_k + xI)^{-1}, C_k^{1/2}, \mathrm{Sq}_k, \qquad x \geq 0,$$

and $(A_k^{\min})^{(2)}(\mathrm{pj}^* A') = \mathrm{pj}^*(A_k^{\min})^{(1)}(A')$. This is exact-operator consistency. It does not assert equality of independently weakened families or the later nonlinear steps.

Proof work. Use common finite row formulas, uniqueness of the Landau minimizer, transpose consistency after canceling J^3, and the resolvent integral for the root. Average an arbitrary larger-torus axial fluctuation over cover translations to verify the pulled-back block minimizer's orthogonality.

Source. QEM1:9010–9069, `eq:photon-exact-cover-consistency` and `eq:photon-minimizer-cover-consistency`. Proof estimate: 0.5–1 page.